

A STABLE-RESIDUAL PRINCIPLE AND AN ALTERNATIVE PROOF OF THE NEGATIVE ANSWER TO ERDŐS PROBLEM 973

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ABSTRACT. We give an alternative proof of the negative answer to Erdős Problem 973. Specifically, for every $\lambda > 0$ there exists an integer $N = N(\lambda)$ such that, for every integer $n \geq N$ and every $z_1, \dots, z_n \in \mathbb{C}$ with $|z_j| \geq 1$,

$$\max_{2 \leq k \leq n+1} \left| \sum_{j=1}^n z_j^k \right| > e^{-\lambda n},$$

even without the normalization $z_1 = 1$. The same conclusion was previously obtained by Luo, Yang, and Zhu. Our contribution is an alternative proof based on the following stable-residual principle. Let r_n be the infimum of

$$\frac{\|P' - P'(0)P\|_{H^2}}{\|P\|_{H^2}}$$

over degree- n polynomials P with $P(0) = 1$ and all zeros in the closed unit disk. We prove that there exist absolute constants $D > 0$ and $N_0 \in \mathbb{N}$ such that

$$r_n \geq \exp\left(-Dn^{2/3}(\log(en))^{2/3}\right) \quad (n \geq N_0),$$

and hence $r_n^{1/n} \rightarrow 1$. An exact projected Newton identity transfers this result to exterior power sums. The residual estimate follows by normalizing at a boundary maximum, integrating a first-order ODE to separate the zeros, and excluding the resulting moment profile with a finite polynomial test.

1. INTRODUCTION

Write $\mathbb{D} = \{w \in \mathbb{C} : |w| < 1\}$. For $z = (z_1, \dots, z_n) \in \mathbb{C}^n$, put

$$p_k(z) = \sum_{j=1}^n z_j^k, \quad M_n(z) = \max_{2 \leq k \leq n+1} |p_k(z)|.$$

Erdős asked whether there is a constant $C > 1$ such that, for every $n \geq 2$, one can choose

$$z_1 = 1, \quad |z_j| \geq 1 \quad (1 \leq j \leq n), \quad M_n(z) < C^{-n};$$

see Erdős [2, p. 213]. The formulation above is the one currently catalogued as Erdős Problem 973 [1]; a related closed-disk formulation appears in Hayman [4, Problem 7.3].

We give an alternative proof that the answer is negative. More precisely, for every $\lambda > 0$, there exists $N = N(\lambda)$ such that, for every $n \geq N$ and every configuration with $|z_j| \geq 1$,

$$M_n(z) > e^{-\lambda n};$$

in particular, the distinguished-point condition $z_1 = 1$ is unnecessary. This quantitative conclusion was obtained earlier by Luo, Yang, and Zhu [5]. The mathematical focus of this paper is the stable-residual theorem below and a self-contained route from it to their exterior-power-sum result. This stable-residual formulation does not appear in [5].

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As noted by Luo–Yang–Zhu, an application of Turán’s first main theorem gives an exponential lower bound with base asymptotic to $2e$; see [6, Theorem 6.1] and [5, Section 1]. An unpublished manuscript of Turturean [7] develops an argument intended to prove

$$M_n(z) \geq \exp(-(0.0597505151 \dots + o(1))n)$$

under the normalization $z_1 = 1$, which would exclude every possible $C > 1.0615716674 \dots$. The cited revision, however, records a remaining gap in the numerical certification of its final bound.

For comparison, under the closed-disk normalization $\max_j |z_j| = 1$, all points lie in $\overline{\mathbb{D}}$ and at least one lies on $\partial\mathbb{D}$. The associated power-sum minimax problem has a genuinely exponential optimum; sharp exponential bounds in that setting were obtained by L. Erdős [3]. For the exterior problem considered here, Luo–Yang–Zhu prove the complete asymptotic statement above by combining a truncated exponential factorization with overconvergence on a fixed open set outside $\overline{\mathbb{D}}$ and a normal-family obstruction for Cauchy transforms.

Notation and norms. Let $\mathbb{T} = \{\zeta \in \mathbb{C} : |\zeta| = 1\}$. For a continuous function f on $\mathbb{T}$, set

$$\|f\|_{L^2(\mathbb{T})} = \left(\frac{1}{2\pi} \int_0^{2\pi} |f(e^{i\theta})|^2 d\theta \right)^{1/2}, \quad \|f\|_{L^\infty(\mathbb{T})} = \max_{|\zeta|=1} |f(\zeta)|.$$

For a polynomial $Q(t) = \sum_{k=0}^m q_k t^k$, define

$$\|Q\|_{H^2} = \left(\sum_{k=0}^m |q_k|^2 \right)^{1/2}.$$

Parseval’s identity gives the exact equality

$$\|Q\|_{H^2} = \|Q\|_{L^2(\mathbb{T})}.$$

Throughout the paper we abbreviate these equal norms by $\|Q\|_2$ and write $\|Q\|_\infty = \|Q\|_{L^\infty(\mathbb{T})}$. By the maximum-modulus principle, $\|Q\|_\infty = \max_{|t| \leq 1} |Q(t)|$.

The main result of this paper isolates a different mechanism. Let $\mathcal{P}_n$ be the class of degree- n polynomials P such that

$$P(0) = 1, \quad \{\text{zeros of } P\} \subseteq \overline{\mathbb{D}}.$$

In this paper, a polynomial is called “Schur-stable” if each of its zeros β satisfies $|\beta| \leq 1$. Thus we use the closed-disk convention, and zeros on $\partial\mathbb{D}$ are allowed. Define

$$\rho(P) = \frac{\|P' - P'(0)P\|_2}{\|P\|_2}.$$

Our main result is a uniform lower bound for this residual.

Theorem 1.1 (Stable-residual principle). *For every $c > 0$, there exists $N(c)$ such that*

$$\rho(P) > e^{-cn}$$

whenever $n \geq N(c)$ and $P \in \mathcal{P}_n$. Moreover, if

$$r_n = \inf_{P \in \mathcal{P}_n} \rho(P),$$

then there exist absolute constants $D > 0$ and $N_0 \in \mathbb{N}$ such that

$$r_n \geq \exp\left(-Dn^{2/3}(\log(en))^{2/3}\right) \quad (n \geq N_0).$$

In particular,

$$\lim_{n \rightarrow \infty} r_n^{1/n} = 1.$$

As an application, an exact projected Newton identity recovers the following theorem of Luo–Yang–Zhu.

Corollary 1.2 (Exterior power sums; Luo–Yang–Zhu). *For every $\lambda > 0$, there exists $N(\lambda)$ such that, for every integer $n \geq N(\lambda)$ and every $z_1, \dots, z_n \in \mathbb{C}$ satisfying $|z_j| \geq 1$ for $1 \leq j \leq n$,*

$$\max_{2 \leq k \leq n+1} \left| \sum_{j=1}^n z_j^k \right| > e^{-\lambda n}.$$

Corollary 1.2 is exactly the stronger, unnormalized conclusion already proved in [5]. We therefore do not claim priority for that theorem or for the negative answer to Problem 973. The distinction lies in the proof architecture: Theorem 1.1 first forgets the individual power sums and studies an arbitrary Schur-stable polynomial with a small H^2 residual. A boundary maximum turns the residual equation into a near-constant-coefficient ODE. Integration from each zero to the maximum gives uniform zero separation. The resulting reciprocal distances have a uniformly positive first moment and exponentially small higher moments. A finite polynomial test shows that these two properties are incompatible.

Section 2 gives the exact reduction from power sums to the stable residual. Sections 3–6 prove Theorem 1.1. Section 7 gives our alternative proof of Corollary 1.2 and its exponential-scale consequences. Section 8 compares the present argument with arXiv:2607.22017 in detail.

2. THE PROJECTED NEWTON IDENTITY

Proposition 2.1. *Let $z_1, \dots, z_n \in \mathbb{C}$ satisfy $|z_j| \geq 1$, and set*

$$p_k = \sum_{j=1}^n z_j^k, \quad M = \max_{2 \leq k \leq n+1} |p_k|, \quad P(t) = \prod_{j=1}^n (1 - z_j t).$$

Then $P \in \mathcal{P}_n$ and

$$\rho(P) \leq nM.$$

Proof. The zeros of P are $1/z_j$, hence lie in $\overline{\mathbb{D}}$, and $P(0) = 1$. Also $P'(0) = -p_1$. Set

$$r_0 = \left(\max_{1 \leq j \leq n} |z_j| \right)^{-1} > 0.$$

If $|t| < r_0$, then $1 - z_j t \neq 0$ for every j , and hence $P(t) \neq 0$. On this disk, the finite geometric identity

$$\frac{z_j}{1 - z_j t} = \sum_{k=1}^{n+1} z_j^k t^{k-1} + \frac{z_j^{n+2} t^{n+1}}{1 - z_j t}$$

gives

$$\begin{aligned} \frac{P'(t)}{P(t)} &= - \sum_{j=1}^n \frac{z_j}{1 - z_j t} \\ &= - \sum_{j=1}^n \left(\sum_{k=1}^{n+1} z_j^k t^{k-1} + \frac{z_j^{n+2} t^{n+1}}{1 - z_j t} \right) \\ &= - \sum_{k=1}^{n+1} \left(\sum_{j=1}^n z_j^k \right) t^{k-1} - t^{n+1} \sum_{j=1}^n \frac{z_j^{n+2}}{1 - z_j t} \\ &= - \sum_{k=1}^{n+1} p_k t^{k-1} - t^{n+1} \sum_{j=1}^n \frac{z_j^{n+2}}{1 - z_j t}. \end{aligned}$$

Multiplying by $P(t)$ and using

$$\frac{P(t)}{1 - z_j t} = \prod_{\substack{1 \leq i \leq n \\ i \neq j}} (1 - z_i t),$$

we obtain

$$P'(t) + P(t) \sum_{k=1}^{n+1} p_k t^{k-1} = -t^{n+1} \sum_{j=1}^n z_j^{n+2} \prod_{\substack{1 \leq i \leq n \\ i \neq j}} (1 - z_i t), \quad |t| < r_0.$$

Both sides are polynomials in t , so the identity holds for every $t \in \mathbb{C}$. Set

$$A(t) = \sum_{k=2}^{n+1} p_k t^{k-1}.$$

Since the right-hand side of the preceding polynomial identity is divisible by t^{n+1} , projection onto degrees at most n gives

$$\begin{aligned} P' &= -\Pi_{\leq n} \left[P \sum_{k=1}^{n+1} p_k t^{k-1} \right] \\ &= -\Pi_{\leq n} (P(p_1 + A)) \\ &= -p_1 P - \Pi_{\leq n} (AP). \end{aligned}$$

Since $P'(0) = -p_1$,

$$(1) \quad P' - P'(0)P = -\Pi_{\leq n} (AP),$$

where $\Pi_{\leq n}$ denotes coefficient projection onto degrees at most n . By Parseval's identity,

$$\|AP\|_2^2 = \frac{1}{2\pi} \int_0^{2\pi} |A(e^{i\theta})P(e^{i\theta})|^2 d\theta \leq \|A\|_\infty^2 \|P\|_2^2.$$

Hence, by (1),

$$\begin{aligned} \rho(P) &= \frac{\|\Pi_{\leq n}(AP)\|_2}{\|P\|_2} \\ &\leq \frac{\|AP\|_2}{\|P\|_2} \\ &\leq \|A\|_\infty \\ &\leq \sum_{k=2}^{n+1} |p_k| \leq nM. \end{aligned}$$

□

Remark 2.2. The reduction is one-way in the form needed here: exponential smallness of all the power sums $p_2, \dots, p_{n+1}$ forces an exponentially small normalized residual. The stable-residual theorem itself is stated for every polynomial in $\mathcal{P}_n$, without a coefficientwise power-sum hypothesis.

3. MAXIMUM-POINT NORMALIZATION

We now prove Theorem 1.1. Suppose to the contrary that there are $c > 0$, integers $n_j \rightarrow \infty$, and $P_j \in \mathcal{P}_{n_j}$ such that

$$(2) \quad \rho_j := \rho(P_j) \leq e^{-cn_j}.$$

Replacing c by $\min(c, 1)$, we may assume $0 < c \leq 1$.

Choose $\tau_j \in \partial\mathbb{D}$ at which $|P_j|$ attains its boundary maximum, and define

$$(3) \quad G_j(s) = \frac{P_j(\tau_j s)}{P_j(\tau_j)}, \quad \mu_j = \tau_j P_j'(0), \quad E_j = G_j' - \mu_j G_j.$$

By the maximum-modulus principle,

$$|P_j(\tau_j)| = \max_{|\zeta|=1} |P_j(\zeta)| \geq |P_j(0)| = 1,$$

and

$$(4) \quad G_j(1) = 1, \quad \|G_j\|_\infty = 1, \quad \|G_j\|_2 \leq \|G_j\|_\infty = 1.$$

Set

$$R_j(t) := P_j'(t) - P_j'(0)P_j(t) = \sum_{k=0}^{n_j} r_{j,k} t^k, \quad P_j(t) = \sum_{k=0}^{n_j} a_{j,k} t^k.$$

Then

$$\begin{aligned} E_j(s) &= G_j'(s) - \mu_j G_j(s) \\ &= \frac{\tau_j P_j'(\tau_j s)}{P_j(\tau_j)} - \frac{\tau_j P_j'(0) P_j(\tau_j s)}{P_j(\tau_j)} \\ &= \frac{\tau_j}{P_j(\tau_j)} (P_j'(\tau_j s) - P_j'(0) P_j(\tau_j s)) \\ &= \frac{\tau_j R_j(\tau_j s)}{P_j(\tau_j)} \\ &= \frac{1}{P_j(\tau_j)} \sum_{k=0}^{n_j} \tau_j^{k+1} r_{j,k} s^k. \end{aligned}$$

Similarly,

$$G_j(s) = \frac{1}{P_j(\tau_j)} \sum_{k=0}^{n_j} \tau_j^k a_{j,k} s^k.$$

By Parseval's identity and $|\tau_j| = 1$,

$$\begin{aligned} \|E_j\|_2^2 &= \frac{\sum_{k=0}^{n_j} |\tau_j^{k+1} r_{j,k}|^2}{|P_j(\tau_j)|^2} = \frac{\|R_j\|_2^2}{|P_j(\tau_j)|^2}, \\ \|G_j\|_2^2 &= \frac{\sum_{k=0}^{n_j} |\tau_j^k a_{j,k}|^2}{|P_j(\tau_j)|^2} = \frac{\|P_j\|_2^2}{|P_j(\tau_j)|^2}. \end{aligned}$$

Hence

$$(5) \quad \frac{\|E_j\|_2}{\|G_j\|_2} = \frac{\|R_j\|_2}{\|P_j\|_2} = \rho(P_j) = \rho_j.$$

Let $\beta_{j,1}, \dots, \beta_{j,n_j}$ be the zeros of G_j , counted with multiplicity. Every $\beta_{j,\ell}$ belongs to $\overline{\mathbb{D}}$, and none equals 1.

Put

$$d_j = G_j'(1).$$

We use the classical Bernstein inequality on the unit circle: if Q has degree at most n , then

$$(6) \quad \|Q'\|_\infty \leq n \|Q\|_\infty, \quad \|Q^{(m)}\|_\infty \leq n(n-1) \cdots (n-m+1) \|Q\|_\infty \quad (1 \leq m \leq n),$$

where the second estimate follows by iteration. Using $G_j(1) = 1$ from (4), we have

$$\frac{d}{d\theta} |G_j(e^{i\theta})|^2 = 2 \operatorname{Re} \left(i e^{i\theta} G_j'(e^{i\theta}) \overline{G_j(e^{i\theta})} \right).$$

The left-hand side vanishes at the maximum point $\theta = 0$. Consequently,

$$0 = \frac{d}{d\theta} |G_j(e^{i\theta})|^2 \Big|_{\theta=0} = 2 \operatorname{Re}(iG'_j(1)) = -2 \operatorname{Im} G'_j(1),$$

and hence

$$(7) \quad d_j \in \mathbb{R}.$$

Since G_j has degree at most n_j , (6) and (4) give

$$(8) \quad |d_j| = |G'_j(1)| \leq \|G'_j\|_\infty \leq n_j \|G_j\|_\infty = n_j.$$

Using $|\tau_j| = 1$, coefficient Cauchy–Schwarz, (4), (2), and (5), we obtain

$$\begin{aligned} |\mu_j - d_j| &= |\mu_j G_j(1) - G'_j(1)| = |E_j(1)| \\ &= \frac{1}{|P_j(\tau_j)|} \left| \sum_{k=0}^{n_j} \tau_j^{k+1} r_{j,k} \right| \\ (9) \quad &\leq \frac{\sqrt{n_j+1}}{|P_j(\tau_j)|} \left(\sum_{k=0}^{n_j} |r_{j,k}|^2 \right)^{1/2} = \sqrt{n_j+1} \|E_j\|_2 \\ &= \sqrt{n_j+1} \rho_j \|G_j\|_2 \\ &\leq \sqrt{n_j+1} e^{-cn_j} \xrightarrow{j \rightarrow \infty} 0. \end{aligned}$$

In particular, $|\mu_j - d_j| < 1$ for all sufficiently large j .

4. THE ODE AND UNIFORM ZERO SEPARATION

Lemma 4.1 (Zero separation). *There exists an index j_0 such that, for every $j \geq j_0$,*

$$1 - \operatorname{Re} \beta_{j,\ell} \geq \frac{c}{2} \quad (1 \leq \ell \leq n_j).$$

Proof. Set $\eta_j = \mu_j - d_j$. By (9), there exists j_1 such that

$$|\eta_j| = |\mu_j - d_j| < 1 \quad (j \geq j_1).$$

Fix $j \geq j_1$ and a zero $\beta = \beta_{j,\ell}$, and put

$$\delta = 1 - \operatorname{Re} \beta.$$

Then $\delta \geq 0$, since $\beta \in \overline{\mathbb{D}}$. The exact differential identity

$$(e^{-\mu_j s} G_j(s))' = e^{-\mu_j s} E_j(s)$$

integrated along the segment from β to 1 yields

$$(10) \quad 1 = \int_{\beta}^1 e^{\mu_j(1-s)} E_j(s) ds.$$

For $|s| \leq 1$, coefficient Cauchy–Schwarz gives

$$(11) \quad |E_j(s)| \leq \sqrt{n_j+1} \|E_j\|_2.$$

Parametrize the segment by

$$s = s(u) = \beta + u(1 - \beta), \quad 0 \leq u \leq 1.$$

Then $1 - s = (1 - u)(1 - \beta)$. Recall that $\mu_j = d_j + \eta_j$, where $d_j \in \mathbb{R}$, $|d_j| \leq n_j$, and $|\eta_j| < 1$. Moreover, $\delta = 1 - \operatorname{Re} \beta \geq 0$, and $|1 - \beta| \leq 1 + |\beta| \leq 2$. Consequently,

$$\begin{aligned}
 \operatorname{Re}(\mu_j(1 - s)) &= (1 - u) \operatorname{Re}(\mu_j(1 - \beta)) \\
 &= (1 - u) [d_j \delta + \operatorname{Re}(\eta_j(1 - \beta))] \\
 (12) \quad &\leq (1 - u) (|d_j| \delta + |\eta_j| |1 - \beta|) \\
 &\leq (1 - u)(n_j \delta + 2) \\
 &\leq n_j \delta + 2.
 \end{aligned}$$

Using $ds = (1 - \beta) du$, (10)–(11) give

$$\begin{aligned}
 1 &\leq |1 - \beta| \int_0^1 e^{\operatorname{Re}(\mu_j(1 - s(u)))} |E_j(s(u))| du \\
 &\leq |1 - \beta| e^{n_j \delta + 2} \sqrt{n_j + 1} \|E_j\|_2.
 \end{aligned}$$

Since

$$\begin{aligned}
 |1 - \beta| &\leq 2, \quad \|E_j\|_2 = \rho_j \|G_j\|_2 \leq \rho_j, \\
 1 &\leq 2e^{n_j \delta + 2} \sqrt{n_j + 1} \rho_j
 \end{aligned}$$

and hence

$$\rho_j \geq \frac{e^{-n_j \delta - 2}}{2\sqrt{n_j + 1}}.$$

Together with $\rho_j \leq e^{-cn_j}$, this gives

$$\delta \geq c - \frac{2 + \log(2\sqrt{n_j + 1})}{n_j}.$$

Since the second term on the right tends to zero, there exists $j_0 \geq j_1$ such that the right-hand side above is at least $c/2$ whenever $j \geq j_0$. Hence, for every $j \geq j_0$ and $1 \leq \ell \leq n_j$,

$$1 - \operatorname{Re} \beta_{j,\ell} = \delta \geq \frac{c}{2}.$$

□

5. RECIPROCAL-ZERO MOMENTS

For each normalized zero, set

$$(13) \quad v_{j,\ell} = \frac{1}{1 - \beta_{j,\ell}}.$$

The disk condition gives

$$(14) \quad \operatorname{Re} v_{j,\ell} = \frac{1 - \operatorname{Re} \beta_{j,\ell}}{|1 - \beta_{j,\ell}|^2} \geq \frac{1}{2},$$

because

$$2(1 - \operatorname{Re} \beta) - |1 - \beta|^2 = 1 - |\beta|^2 \geq 0.$$

Lemma 4.1 gives, for $j \geq j_0$,

$$(15) \quad |v_{j,\ell}| \leq \frac{2}{c}.$$

For $k \geq 1$, set

$$m_{j,k} = \frac{1}{n_j} \sum_{\ell=1}^{n_j} v_{j,\ell}^k.$$

Factoring

$$G_j(s) = C_j \prod_{\ell=1}^{n_j} (s - \beta_{j,\ell})$$

and using $G_j(1) = 1 \neq 0$, we obtain, in a neighborhood of 1,

$$\frac{G'_j(s)}{G_j(s)} = \sum_{\ell=1}^{n_j} \frac{1}{s - \beta_{j,\ell}}.$$

At $s = 1$, (7) and (14) give

$$(16) \quad m_{j,1} = \frac{G'_j(1)}{n_j G_j(1)} = \frac{d_j}{n_j} \in \mathbb{R}, \quad m_{j,1} \geq \frac{1}{2}.$$

In a neighborhood of 1, define

$$F_j(s) = \frac{G'_j(s)}{G_j(s)} - \mu_j = \sum_{\ell=1}^{n_j} \frac{1}{s - \beta_{j,\ell}} - \mu_j = \frac{E_j(s)}{G_j(s)}.$$

Since $E_j = F_j G_j$, differentiation m times and evaluation at 1 give

$$(17) \quad E_j^{(m)}(1) = \sum_{r=0}^m \binom{m}{r} F_j^{(r)}(1) G_j^{(m-r)}(1).$$

For $0 \leq m \leq n_j$, (6), coefficient Cauchy–Schwarz, (4), and (5) give

$$(18) \quad \begin{aligned} |E_j^{(m)}(1)| &\leq \|E_j^{(m)}\|_{\infty} \leq n_j^m \|E_j\|_{\infty} \leq n_j^m \sqrt{n_j + 1} \rho_j, \\ |G_j^{(m)}(1)| &\leq \|G_j^{(m)}\|_{\infty} \leq n_j^m. \end{aligned}$$

Define constants recursively by

$$A_0 = 1, \quad A_m = 1 + \sum_{r=0}^{m-1} \binom{m}{r} A_r \quad (m \geq 1).$$

For $0 \leq m \leq n_j$, induction on m gives

$$(19) \quad |F_j^{(m)}(1)| \leq A_m n_j^m \sqrt{n_j + 1} \rho_j$$

for every j . The case $m = 0$ follows from $F_j(1) = E_j(1)$; for $m \geq 1$, (17) gives

$$\begin{aligned} |F_j^{(m)}(1)| &\leq |E_j^{(m)}(1)| + \sum_{r=0}^{m-1} \binom{m}{r} |F_j^{(r)}(1)| |G_j^{(m-r)}(1)| \\ &\leq \left(1 + \sum_{r=0}^{m-1} \binom{m}{r} A_r \right) n_j^m \sqrt{n_j + 1} \rho_j \\ &= A_m n_j^m \sqrt{n_j + 1} \rho_j. \end{aligned}$$

We claim that

$$(20) \quad A_m \leq 3^m m! \quad (m \geq 0).$$

Indeed, the case $m = 0$ is immediate. Let $m \geq 1$, and suppose that $A_r \leq 3^r r!$ for $0 \leq r < m$. Then

$$\begin{aligned} A_m &\leq 1 + \sum_{r=0}^{m-1} \binom{m}{r} 3^r r! = 1 + m! \sum_{r=0}^{m-1} \frac{3^r}{(m-r)!} \\ &= 1 + m! \sum_{q=1}^m \frac{3^{m-q}}{q!} \\ &= 3^m m! \left(\frac{1}{3^m m!} + \sum_{q=1}^m \frac{1}{3^q q!} \right) \\ &\leq 3^m m! \left(\frac{1}{3} + \sum_{q=1}^{\infty} \frac{1}{3^q} \right) = \frac{5}{6} 3^m m! < 3^m m!. \end{aligned}$$

For $2 \leq k \leq n_j + 1$, differentiating the partial-fraction expression for F_j $k-1$ times eliminates the constant $-\mu_j$. Evaluating at $s = 1$ gives

$$\begin{aligned} F_j^{(k-1)}(1) &= \sum_{\ell=1}^{n_j} \frac{d^{k-1}}{ds^{k-1}} \frac{1}{s - \beta_{j,\ell}} \Big|_{s=1} \\ (21) \quad &= (-1)^{k-1} (k-1)! \sum_{\ell=1}^{n_j} \frac{1}{(1 - \beta_{j,\ell})^k} \\ &= (-1)^{k-1} (k-1)! \sum_{\ell=1}^{n_j} v_{j,\ell}^k. \end{aligned}$$

By (2), (19), (20), and (21), for $2 \leq k \leq n_j + 1$,

$$\begin{aligned} (22) \quad |m_{j,k}| &= \frac{|F_j^{(k-1)}(1)|}{n_j(k-1)!} \\ &\leq \frac{A_{k-1}}{(k-1)!} n_j^{k-2} \sqrt{n_j + 1} \rho_j \\ &\leq 3^{k-1} n_j^{k-2} \sqrt{n_j + 1} e^{-cn_j}. \end{aligned}$$

6. A FINITE POLYNOMIAL MOMENT CONTRADICTION

For $j \geq j_0$, (14) and (15) give

$$\operatorname{Re} v_{j,\ell} \geq \frac{1}{2}, \quad |v_{j,\ell}| \leq \frac{2}{c}.$$

Set

$$R = \frac{8}{c^2}, \quad \vartheta = \sqrt{1 - \frac{c^2}{16}}.$$

Because $0 < c \leq 1$, we have $R \geq 8$ and $0 < \vartheta < 1$. Moreover, for every $v = v_{j,\ell}$,

$$\begin{aligned} \left| 1 - \frac{v}{R} \right|^2 &= 1 - \frac{2 \operatorname{Re} v}{R} + \frac{|v|^2}{R^2} \\ &\leq 1 - \frac{c^2}{8} + \frac{c^2}{16} = \vartheta^2. \end{aligned}$$

For every j , define

$$K_j = \left\lceil \frac{cn_j}{2 \log(6n_j)} \right\rceil.$$

Since $n_j \rightarrow \infty$,

$$K_j \rightarrow \infty, \quad \frac{K_j}{n_j} \rightarrow 0.$$

Therefore, there exists $j_2 \geq j_0$ such that $2 \leq K_j \leq n_j$ for every $j \geq j_2$. For such j , set

$$\Phi_j(v) = v \left(1 - \frac{v}{R}\right)^{K_j-1}.$$

$$(23) \quad \begin{aligned} \left| \frac{1}{n_j} \sum_{\ell=1}^{n_j} \Phi_j(v_{j,\ell}) \right| &\leq \frac{1}{n_j} \sum_{\ell=1}^{n_j} |v_{j,\ell}| \left| 1 - \frac{v_{j,\ell}}{R} \right|^{K_j-1} \\ &\leq \frac{1}{n_j} \sum_{\ell=1}^{n_j} \frac{2}{c} \vartheta^{K_j-1} = \frac{2}{c} \vartheta^{K_j-1} \rightarrow 0 \quad (j \rightarrow \infty). \end{aligned}$$

On the other hand, the binomial expansion gives

$$\Phi_j(v) = v + \sum_{k=2}^{K_j} a_{j,k} v^k, \quad a_{j,k} = (-1)^{k-1} \binom{K_j-1}{k-1} R^{-(k-1)}.$$

Since $R \geq 8$,

$$(24) \quad |a_{j,k}| \leq 2^{K_j-1} \quad (2 \leq k \leq K_j).$$

By the definition of K_j ,

$$(K_j - 1) \log(6n_j) \leq \frac{cn_j}{2}, \quad (6n_j)^{K_j-1} \leq e^{cn_j/2}.$$

For $j \geq j_2$, (22) and (24) give

$$(25) \quad \begin{aligned} \left| \sum_{k=2}^{K_j} a_{j,k} m_{j,k} \right| &\leq \sum_{k=2}^{K_j} 2^{K_j-1} 3^{k-1} n_j^{k-2} \sqrt{n_j+1} e^{-cn_j} \\ &\leq (K_j - 1) 2^{K_j-1} 3^{K_j-1} n_j^{K_j-2} \sqrt{n_j+1} e^{-cn_j} \\ &= \frac{K_j - 1}{n_j} (6n_j)^{K_j-1} \sqrt{n_j+1} e^{-cn_j} \\ &\leq \frac{K_j - 1}{n_j} \sqrt{n_j+1} e^{-cn_j/2} \\ &\leq \sqrt{n_j+1} e^{-cn_j/2} \rightarrow 0 \quad (j \rightarrow \infty). \end{aligned}$$

Therefore, averaging the binomial identity for Φ_j over $v_{j,1}, \dots, v_{j,n_j}$ gives

$$\begin{aligned} \frac{1}{n_j} \sum_{\ell=1}^{n_j} \Phi_j(v_{j,\ell}) &= \frac{1}{n_j} \sum_{\ell=1}^{n_j} \left(v_{j,\ell} + \sum_{k=2}^{K_j} a_{j,k} v_{j,\ell}^k \right) \\ &= m_{j,1} + \sum_{k=2}^{K_j} a_{j,k} m_{j,k}. \end{aligned}$$

Together with (23) and (25), this yields

$$m_{j,1} = \frac{1}{n_j} \sum_{\ell=1}^{n_j} \Phi_j(v_{j,\ell}) - \sum_{k=2}^{K_j} a_{j,k} m_{j,k} \rightarrow 0 \quad (j \rightarrow \infty),$$

contradicting (16). This proves the asserted lower bound in Theorem 1.1.

For the quantitative bound, fix $0 < c \leq 1$ and $n \geq 2$, and suppose for contradiction that some $P \in \mathcal{P}_n$ satisfies $\rho(P) \leq e^{-cn}$. Repeat the preceding normalization with the subscript j omitted, and set

$$K = \left\lceil \frac{cn}{2 \log(6n)} \right\rceil.$$

The contradiction follows provided that

$$(26) \quad \begin{aligned} \sqrt{n+1}e^{-cn} &< 1, & \frac{2 + \log(2\sqrt{n+1})}{n} &\leq \frac{c}{2}, & 2 \leq K \leq n, \\ \frac{2}{c}\vartheta^{K-1} &\leq \frac{1}{4}, & \sqrt{n+1}e^{-cn/2} &< \frac{1}{4}. \end{aligned}$$

Indeed, the first condition gives $|\mu - d| < 1$ by (9); the second gives $1 - \operatorname{Re} \beta_\ell \geq c/2$ by the calculation in Lemma 4.1; and the third permits the use of (22) for every $2 \leq k \leq K$. The last two conditions, together with (16), give

$$\begin{aligned} \frac{1}{2} &\leq m_1 = |m_1| \\ &\leq \left| \frac{1}{n} \sum_{\ell=1}^n \Phi(v_\ell) \right| + \left| \sum_{k=2}^K a_k m_k \right| \\ &\leq \frac{2}{c} \vartheta^{K-1} + \sqrt{n+1}e^{-cn/2} < \frac{1}{2}, \end{aligned}$$

which is impossible.

It remains to give a single sufficient condition for (26). Since

$$\vartheta = \sqrt{1 - \frac{c^2}{16}} \leq e^{-c^2/32},$$

if

$$\frac{cn}{2 \log(6n)} \geq 2, \quad \frac{c^3 n}{128 \log(6n)} \geq \log \frac{8}{c},$$

then

$$K - 1 \geq \frac{cn}{4 \log(6n)}, \quad \frac{2}{c} \vartheta^{K-1} \leq \frac{2}{c} \exp\left(-\frac{c^3 n}{128 \log(6n)}\right) \leq \frac{1}{4}.$$

Put $C_0 = 10^4$, $L = \log(e/c)$, and $n_* = C_0 c^{-3} L^2$. Since $L \geq 1$,

$$\log(6n_*) = \log(60000) + 3(L-1) + 2 \log L < 17L.$$

The function $x/\log(6x)$ is increasing for $x \geq 1$. Hence, for every $n \geq n_*$,

$$\frac{c^3 n}{128 \log(6n)} > \frac{10^4}{128 \cdot 17} L > 3L \geq \log \frac{8}{c}, \quad \frac{cn}{2 \log(6n)} > \frac{10^4}{34} c^{-2} L > 2.$$

In particular, $cn \geq 10^4$, $n \geq 10^4$, and $cn \geq 4 \log(6n)$. Therefore

$$\sqrt{n+1}e^{-cn/2} \leq e^{-3cn/8} < \frac{1}{4},$$

and

$$\frac{2 + \log(2\sqrt{n+1})}{n} \leq \frac{\log(6n)}{n} \leq \frac{c}{4}, \quad 2 \leq K \leq \left\lceil \frac{n}{2} \right\rceil \leq n.$$

The first condition in (26) follows from the first of these estimates. Thus all the conditions in (26) hold whenever

$$(27) \quad 0 < c \leq 1, \quad n \geq C_0 c^{-3} \log^2\left(\frac{e}{c}\right).$$

Consequently, no $P \in \mathcal{P}_n$ can satisfy $\rho(P) \leq e^{-cn}$, and hence

$$r_n \geq e^{-cn}.$$

Choose an absolute constant $D > 0$ such that

$$\frac{C_0}{9D^3} < 1,$$

and set

$$c_n = Dn^{-1/3}(\log(en))^{2/3}.$$

Then

$$c_n \longrightarrow 0, \quad \frac{\log(e/c_n)}{\log(en)} \longrightarrow \frac{1}{3},$$

and therefore

$$\frac{C_0 c_n^{-3} \log^2(e/c_n)}{n} = \frac{C_0}{D^3} \left(\frac{\log(e/c_n)}{\log(en)} \right)^2 \longrightarrow \frac{C_0}{9D^3} < 1.$$

Hence there exists an integer N_0 such that, for every $n \geq N_0$,

$$0 < c_n \leq 1, \quad n \geq C_0 c_n^{-3} \log^2\left(\frac{e}{c_n}\right).$$

Applying (27) with $c = c_n$ gives

$$r_n \geq e^{-c_n n} = \exp\left(-Dn^{2/3}(\log(en))^{2/3}\right) \quad (n \geq N_0).$$

Finally, take $P(t) = 1 - t^n$ for $n \geq 2$. Then $P \in \mathcal{P}_n$, $P'(0) = 0$, and

$$\rho(P) = \frac{n}{\sqrt{2}}.$$

Thus $r_n \leq n/\sqrt{2}$, so $\limsup_{n \rightarrow \infty} r_n^{1/n} \leq 1$. The lower bound just proved gives $\liminf_{n \rightarrow \infty} r_n^{1/n} \geq e^{-c}$ for every $c > 0$. Letting $c \downarrow 0$ proves $\lim_{n \rightarrow \infty} r_n^{1/n} = 1$, and completes the proof.

7. APPLICATION: AN ALTERNATIVE PROOF FOR EXTERIOR POWER SUMS

Proof of Corollary 1.2. Fix $\lambda > 0$ and put $c = \lambda/2$. Suppose the conclusion fails for infinitely many n . For those n , choose a configuration with $|z_j| \geq 1$ and

$$M_n(z) \leq e^{-\lambda n}.$$

Proposition 2.1 gives a polynomial $P \in \mathcal{P}_n$ such that

$$\rho(P) \leq ne^{-\lambda n}.$$

For all sufficiently large n , Theorem 1.1 and $\log n < \lambda n/2$ give the contradiction

$$e^{-cn} < \rho(P) \leq ne^{-\lambda n} < e^{-cn}.$$

□

Corollary 7.1 (Problem 973). *There is no constant $C > 1$ such that, for every $n \geq 2$, one can choose $z_1 = 1$, $|z_j| \geq 1$ for $1 \leq j \leq n$, and*

$$\max_{2 \leq k \leq n+1} \left| \sum_{j=1}^n z_j^k \right| < C^{-n}.$$

Proof. Apply Corollary 1.2 with $\lambda = \log C$. The corollary excludes the required configuration for every sufficiently large n , even without imposing $z_1 = 1$. □

Define

$$\Delta_n = \inf_{\substack{z \in \mathbb{C}^n \\ |z_j| \geq 1}} M_n(z), \quad \Delta_n^* = \inf_{\substack{z \in \mathbb{C}^n \\ z_1=1, |z_j| \geq 1}} M_n(z).$$

Corollary 7.2 (Optimal exponential scale).

$$\lim_{n \rightarrow \infty} \Delta_n^{1/n} = \lim_{n \rightarrow \infty} (\Delta_n^*)^{1/n} = 1.$$

Proof. For every $\lambda > 0$, Corollary 1.2 gives $\Delta_n, \Delta_n^* \geq e^{-\lambda n}$ for all sufficiently large n . Taking $z_1 = \cdots = z_n = 1$ gives $\Delta_n \leq \Delta_n^* \leq n$. Hence, for every $\lambda > 0$ and all sufficiently large n ,

$$e^{-\lambda} \leq \Delta_n^{1/n} \leq (\Delta_n^*)^{1/n} \leq n^{1/n}.$$

First let $n \rightarrow \infty$, and then let $\lambda \downarrow 0$. □

8. COMPARISON WITH LUO–YANG–ZHU

This section records precisely what is shared with, and what differs from, the proof in Luo–Yang–Zhu [5]. Its purpose is both mathematical and bibliographic: the exterior-power-sum conclusion has priority in that paper, while the present manuscript offers a different route to it.

8.1. Identical theorem-level conclusion. Theorem 1.1 of [5] and Corollary 1.2 above have the same quantifiers and the same strict exponential lower bound:

$$\begin{aligned} \forall \lambda > 0, \quad \exists N(\lambda) \in \mathbb{N}, \quad \forall n \geq N(\lambda), \quad \forall z = (z_1, \dots, z_n) \in \mathbb{C}^n \text{ with} \\ |z_j| \geq 1 \quad (1 \leq j \leq n), \quad M_n(z) > e^{-\lambda n}. \end{aligned}$$

Both remove the distinguished-point condition $z_1 = 1$, both imply Corollary 7.1, and both yield the two n -th-root limits in Corollary 7.2. There is therefore no difference in the strength of the final exterior-power-sum theorem.

8.2. The Luo–Yang–Zhu mechanism. Under a contradictory bound

$$|p_k| \leq e^{-\lambda n}, \quad 2 \leq k \leq n+1,$$

Luo–Yang–Zhu work directly with

$$F_n(t) = \prod_{j=1}^n (1 - z_j t), \quad x_n = -p_1.$$

They introduce

$$D_n(t) = \sum_{k=2}^{n+1} \frac{p_k}{k} t^k, \quad A_n(t) = e^{-D_n(t)},$$

and use the formal factorization

$$F_n(t) \equiv e^{x_n t} A_n(t) \pmod{t^{n+2}}.$$

Coefficient majorants and the vanishing coefficient at degree $n+1$ control x_n/n . After taking a subsequence,

$$\frac{x_n}{n} \longrightarrow c, \quad 0 < |c| < 1,$$

and, writing $r = |c|$, their endpoint estimate gives

$$\lambda \leq r - 1 - \log r.$$

They then choose a closed disk $K \subseteq \{|w| > 1\}$ on which the truncated exponential is in its convergent regime while the coefficient perturbation remains exponentially negligible. Uniformly on K ,

$$F_n(w) = e^{x_n w} (1 + o(1)), \quad \frac{F'_n(w)}{n F_n(w)} \longrightarrow c \neq 0.$$

Writing $\alpha_{j,n} = z_{j,n}^{-1} \in \overline{\mathbb{D}}$, the last quantity is the Cauchy transform

$$\frac{1}{n} \sum_{j=1}^n \frac{1}{w - \alpha_{j,n}}.$$

Normality on $\mathbb{C} \setminus \overline{\mathbb{D}}$, the identity theorem, and decay at infinity force the constant limit c to be zero, a contradiction.

8.3. The stable-residual mechanism here. The present proof separates the power-sum input from the analytic contradiction. Proposition 2.1 turns small power sums into the single norm inequality

$$\rho(P) \leq nM_n(z).$$

The proof then works with an arbitrary $P \in \mathcal{P}_n$ having exponentially small $\rho(P)$; no individual p_k , no asymptotic scale for p_1 , and no truncated exponential appear.

The essential steps are instead:

- (i) normalize P at a boundary maximum so that $G(1) = \|G\|_\infty = 1$;
- (ii) integrate the exact ODE $(e^{-\mu s} G(s))' = e^{-\mu s} E(s)$ from each zero to 1, obtaining uniform separation of every zero from the maximum point;
- (iii) form the reciprocal distances $v = (1 - \beta)^{-1}$, whose real parts are at least $1/2$;
- (iv) estimate derivatives of E/G uniformly up to order $K \asymp n/\log n$, making the corresponding higher moments exponentially small while the first moment is at least $1/2$;
- (v) apply the finite polynomial test $v(1 - v/R)^{K-1}$, whose average is both small by contraction and close to the first moment by the higher-moment estimates.

The comparison may be summarized as follows.

Feature	Luo–Yang–Zhu [5]	Present proof
Initial analytic object	Individual power sums and a truncated exponential factorization	H^2 norm of $P' - P'(0)P$
Uncontrolled linear term	Proves $x_n/n \rightarrow c \neq 0$ along a subsequence	Absorbed into the ODE parameter μ
Decisive estimate	Overconvergence on an open disk outside $\overline{\mathbb{D}}$	ODE separation of every zero from a boundary maximum
Limit process	Normal-family subsequence outside $\overline{\mathbb{D}}$	No normal-family or limiting-measure argument; a finite- n quantitative contradiction is also available
Final test	Exterior Cauchy transform on $\mathbb{C} \setminus \overline{\mathbb{D}}$	Polynomial $v(1 - v/R)^{K-1}$ on the reciprocal distances
Final obstruction	A nonzero constant exterior Cauchy-transform limit must vanish	The test average tends to zero while its linear term stays at least $1/2$

8.4. Shared skeleton and exact priority statement. The two arguments are not unrelated. Both encode the exterior points as zeros in $\overline{\mathbb{D}}$, and both use logarithmic derivatives to extract information about those zeros. They diverge at the final obstruction. Luo–Yang–Zhu use exterior overconvergence, a normal-family limit, and a Cauchy transform. The present proof uses maximum-point ODE separation and a finite polynomial test, without a limiting measure or analytic continuation.

Thus the appropriate claims are:

- the negative answer to Erdős Problem 973, removal of $z_1 = 1$, and the two exterior-power-sum limits in Corollary 7.2 are not new here;
- the present paper supplies a self-contained alternative proof;
- the stable-residual statement in Theorem 1.1 does not appear in [5];
- no claim of theorem priority over arXiv:2607.22017 is made.

9. FURTHER QUESTIONS

The proof of Theorem 1.1 gives absolute constants $D > 0$ and $N_0 \in \mathbb{N}$ such that

$$\inf_{P \in \mathcal{P}_n} \frac{\|P' - P'(0)P\|_2}{\|P\|_2} \geq \exp\left(-Dn^{2/3}(\log(en))^{2/3}\right) \quad (n \geq N_0).$$

It remains natural to determine the finer order of

$$\inf_{P \in \mathcal{P}_n} \frac{\|P' - P'(0)P\|_2}{\|P\|_2}.$$

A polynomial lower bound, a matching construction, or an improved subexponential exponent would sharpen the finite polynomial-test argument. Other natural variants replace H^2 by H^p , use weighted coefficient norms, or constrain the zeros to a smaller disk.

Likewise, Corollary 1.2 determines the exponential scale of Δ_n and Δ_n^* but not their finer asymptotic order. The present proof does not decide whether these quantities remain bounded away from zero or decay at a polynomial or another subexponential rate.

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DISCLOSURE OF AUTOMATED ASSISTANCE

OpenAI GPT-5.6 Sol was used during proof exploration and drafting; other generative AI systems were used for adversarial checking, literature comparison, and language editing. The argument in this paper is self-contained, and no model output is used as a mathematical premise. The named authors assume full responsibility for the final manuscript, including every statement, proof, citation, and priority claim.

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